Multi-master CAN

Minimum FW version required

The following firmware versions (or later) are required to ensure compatibility with the Multi-master CAN update for supported devices:

Nexus ECU: 1.28.x
Elite ECU: 3.12.x
Nexus PD16: 1.09.x
CAN Keypad: All keypads sold by Haltech
Haltech Dashes (uC-10 and iC-7): 2.28.x
NSP software version: 1.40.x

Overview

Haltech CAN devices, such as the ECU, Dash Display, PD16, and CAN Keypads, are given increased flexibility with this firmware update. Previously, a PD16 or CAN Keypad could only be assigned to be controlled by the ECU.



Before Multi-master CAN implementation

Now, the same PD16 and CAN Keypad can be shared between the ECU and the Dash Display, enabling more dynamic system setups like those shown below. In some applications, an ECU is not even required, as a PD16 or CAN Keypad can be operated using only a Haltech Dash Display. This change transforms the Haltech CAN system into a multi-master CAN network, allowing multiple controllers to access and manage the same devices.

Consequently, users have greater flexibility and customization options for controlling various components in their vehicle.



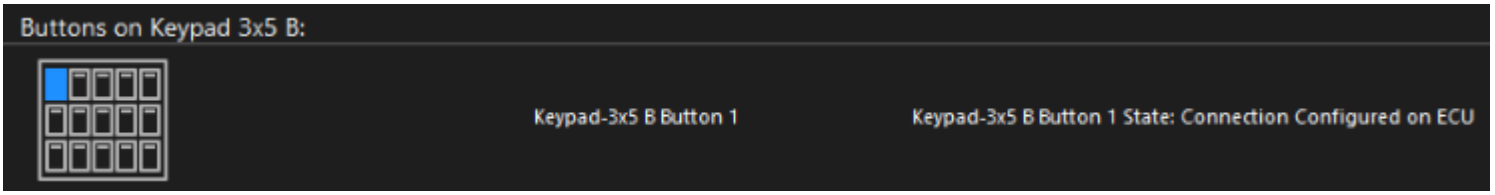
Examples of CAN setups after the Multi-master CAN update

Configuration changes with Multi-master CAN network

With the implementation of this update, devices like CAN Keypads and PD16s are now treated as collections of buttons or IO elements (e.g., buttons, AVIs, SPIs, HCOs, etc.) rather than as individual devices. ECUs and Dash Displays can independently utilize different buttons from the same keypad or separate IOs on the same PD16. However, they cannot share the exact same buttons or IOs.

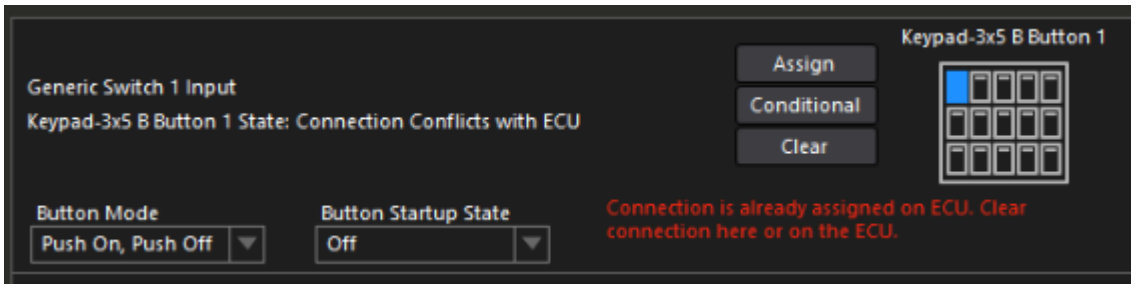


In the example above, the desired CAN keypad buttons are assigned by first connecting the NSP software to the ECU and assigning them. Then, the Dash display is connected to NSP, and the rest of the buttons are assigned to the Dash display to complete the setup. During this process, the NSP software will show buttons (or IOs) that are already assigned to the ECU (i.e., the first master) when assigning the remaining buttons to the Dash display (i.e., the second master). This streamlined process simplifies configuration and helps prevent confusion when assigning buttons or IOs. However, it requires all devices to be online.



Assignment Conflict Handling

When an assignment conflict occurs, the NSP software will generate a DTC on the ECU, highlight the function red in the navigation tree, and display the error message shown below. If a re-assignment is needed, the NSP software provides the option to clear the assignment on the master device you are currently connected to. For example, if a connection is assigned to both the ECU and the Dash display, and you clear it on the ECU, the assignment will remain on the Dash display.



- For Keypad button conflicts, the button's error LED will be triggered. The error LED is the third one, which lights up red, as shown below. Additionally, with this update, the Keypad LEDs will not illuminate when a button is pressed unless that button is assigned to a function. However, the NSP software will still display the button state even if some buttons are left unassigned.

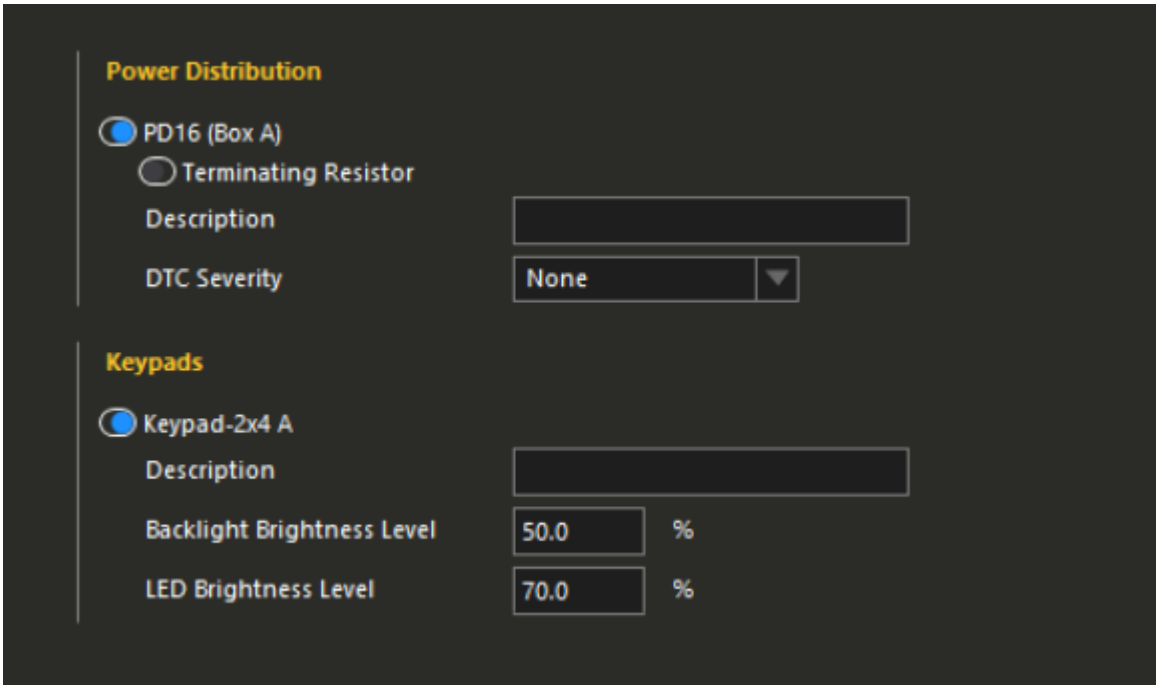


- If a PD16 output is in conflict, it will enter the safe state. The safe state is a user-defined setting in the NSP software for PD16 outputs, which determines the output's behavior if the PD16 loses its CAN connection. In this multi-master CAN update, the safe state will be based on the configuration set by the first master on the bus. The first master is the device that boots up first in the CAN network. Typically, between an ECU and a Dash display, the ECU is the first master, as it generally boots up faster than the Dash display.

Global Device Settings

The following settings are considered to be global settings across PD16s and CAN Keypads:

PD16 - Terminating Resistor
CAN Keypad - Backlight Brightness Level
CAN Keypad - LED Brightness Level



When any of the above settings are modified whilst online, the changes are propagated across the bus to all master devices. For example, if the LED brightness is adjusted to 80% on the Dash display, and then the ECU is connected to NSP, the LED brightness will reflect the updated 80% setting.

If any of the settings are changed while a master device is offline, the last master device to rejoin the network will dictate the values for all devices.

Note: “Description” settings are not considered global settings. This means a CAN Keypad or PD16 may be named differently on the Dash display compared to the ECU.